Assignment – 1.7

IP addressing & Subnetting

Q - 1. List the IP address classes (A, B, C, D, E) and give one real-world example of a device or service that might use each class. Write your answers in a table format.

IP Class	IP Address Range	Use	Example
Class A	1–126	Large networks	Large company
Class B	128–191	Medium networks	University
Class C	192–223	Small networks	Home Wi-Fi
Class D	224–239	Multicast communication	Video streaming
Class E	240–255	Experimental use	Research lab

Q - 2. Given the IP address 192.168.10.45 with subnet mask 255.255.255.0, calculate the network address, broadcast address, and the range of valid host addresses. Hint: Show your working steps for each calculation.

➢ Given:

- **IP Address:** 192.168.10.45
- **Subnet Mask:** 255.255.255.0

Calculation	Answer
1. Network Address	192.168.10.0
2. Broadcast Address	192.168.10.255
3. First Valid Host	192.168.10.1
4. Last Valid Host	192.168.10.254
5. Total Valid Hosts	254

➢ Working Steps:

1. **Network Address:** Set the last octet to 0 → 192.168.10.0
2. **Broadcast Address:** Set the last octet to 255 → 192.168.10.255
3. **Valid Host Range:** From 192.168.10.1 to 192.168.10.254.

<u>Q - 3.</u> Create a subnetting plan for a small food delivery startup like Zomato that needs 4 separate subnets: one for Delivery Partners, one for Restaurants, one for Customers, and one for Admins. Assign a unique subnet (with network address and subnet mask) for each group using the 10.0.0.0/24 network.

➢ Subnetting Plan for a Food Delivery Startup:

- Network: 10.0.0.0/24
- We can divide this network into 4 separate subnets using /26 subnet masks.

Group	Network Address	Subnet Mask
Delivery Partners	10.0.0.0/26	255.255.255.192
Restaurants	10.0.0.64/26	255.255.255.192
Customers	10.0.0.128/26	255.255.255.192
Admins	10.0.0.192/26	255.255.255.192

➢ Result:

- Total subnets: 4
- Usable host addresses per subnet: 62
- Each group has a separate subnet for better network management and security.

Q - 4. You are given the following scenario: A college WiFi network uses the IP range 172.16.0.0/16. The admin wants to create at least 20 subnets for different buildings. Calculate the new subnet mask and the number of hosts available per subnet. Constraint: Show your calculation steps and final answers.

➢ Subnetting Calculation:

Given:

- IP range: 172.16.0.0/16
- Required subnets: 20

Step 1: Bits required

$$2^5 = 32$$

We need 5 bits to create at least 20 subnets.

Step 2: New subnet mask

$$16 + 5 = 21$$

New subnet mask: **255.255.248.0 (/21)**

Step 3: Number of subnets

$$2^5 = 32 \text{ subnets}$$

Step 4: Hosts per subnet

$$32 - 21 = 11 \text{ host bits}$$
$$2^{11} - 2 = 2046$$

➢ Final Answer:

- **New subnet mask:** 255.255.248.0 (/21)
- **Total subnets:** 32
- **Usable hosts per subnet:** 2046

<u>Q - 5.</u> A friend sends you a message asking why their device with IP 192.168.1.300 is not connecting to the network. Write a short explanation identifying the error and what a valid IP address in that subnet should look like.

- The IP address **192.168.1.300 is invalid** because each part of an IPv4 address must be between **0 and 255**. The number 300 is too large.

- **Correct example:** 192.168.1.100

- A valid IP address in the 192.168.1.x subnet has the first three parts as 192.168.1 and the last part between 1 and 254 $with subnet mask$‘255.255.255.0‘.